constituents, instead of approaching every object by the same universal number system developed a priori and independently of the applications" (quoted in [**52**, p. 240]). In his 1928 book, *The Theory of Groups and Quantum Mechanics*, Weyl argued that the essence of symmetries central to quantum mechanics was to be found not in the continuum of real numbers but in the concepts of group theory, particularly in the "reciprocity" between the representations of symmetric permutation groups and complete linear groups [**60**, p. vii].

The physics community welcomed the help of a mathematician with cries of outrage and disgust. The American theoretical physicist John Slater christened the approach of Weyl and his followers "Gruppenpest", or "the pest of group theory". "The authors of the 'Gruppenpest' wrote papers which were incomprehensible to those like me who had not studied group theory", he later confessed. Their results appeared "negligible" to him, and he widely shared his "frustrating experience, worthy of the name of a pest", with other physicists, who were, as he claimed, "as disgusted as I had been with the group-theoretical approach to the problem" (quoted in [**55**, p. x]). As late as 1975, he still believed that he had "slain the 'Gruppenfest'", blissfully unaware of the essential and now routine applications of group theory in elementary particle physics.

More sympathetic physicists also had difficulty grappling with Weyl's mathematics, but this only heightened their admiration for him. Julian Schwinger had to admit that he had not "ever — not even to this day — fully mastered" Weyl's *Theory of Groups and Quantum Mechanics*. Yet he called Weyl "one of my gods", explaining that "the ways of gods are mysterious, inscrutable, and beyond the comprehension of ordinary mortals". When asked if he ever met a scholar he could not understand, Paul Dirac unhesitatingly replied, "Weyl". Yet he found Weyl's — purely mathematical — argument that electrons and antielectrons must have the same mass so disarmingly convincing that he reportedly derived from this encounter his famous maxim, "[I]t is more important to have beauty in one's equations than to have them fit experiment" (quoted in [**42**, p. 11, 19, 12]). Weyl expressed a similar sentiment with respect to mathematics: "My work has always tried to unite the true with the beautiful,